

Statistical measures for measurement scales

(Completed)

① Measures of central tendency:

Mean

- Arithmetic
- Geom ()
- Harmonic mean

Median

Mode

Trimmed - ~~mean~~ ^{mean} weighted after removing the extremes: $x_i, i > 1 \dots n$

$$\frac{\sum w_i x_i}{\sum w_i}$$

1 2 4 6

	sort	exp	variants
Mean	x	$\sum \frac{x_i}{n}$	Trimmed - mean ^{mean} weighted after removing the extremes: $x_i, i > 1 \dots n$
Median	✓	$\uparrow$ n-odd $x_{(\frac{n+1}{2})}$ n-even	
Mode	✓	$\frac{1}{2} (x_{\frac{n}{2}} + x_{\frac{n}{2}+1})$ mode	

②

Measures of Dispersion / Variation

$x_1, \dots, x_n$

Range: —

$$x_H - x_L$$

x_H — max^m Score

Variance —

popⁿ

$$\sum_{i=1}^N (x_i - \bar{x})^2$$

mean x_L — min^m Score

Standard deviation —

Sample —

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

variance

Quartile Deviation

Percentiles

Quartile
quantile

decile :-

Percentile position of x_i

$P(x_i)$

$$= \frac{(i-1) \times 100}{(n-1)}$$

$$\frac{i}{n} \times 100$$

2, 3, 5, 6, 8, 10, 12, 15, 18, 20

_{1 2 3 4 5 6 7 8 9 10}

$$n = 10$$

$$x_i = 12$$

$$P(x_i)$$

$$= \frac{i-1}{n-1} \times 100 =$$

$$= \frac{7-1}{10-1} \times 100 = \frac{6}{9} \times 100 = 67\%$$

Q1

What is

Q2:

Find

the 25th percentile

$$P_{25} = \frac{P \times (n-1)}{100} =$$

$$= \frac{25 \times 9}{100}$$

$$\frac{9}{4} = 2.25$$

$i = 1 = 2.25$
 $i = 3.25$

$$i-1$$

$$i-1 = \frac{P(x_i) \times (n-1)}{100} =$$